KG COLLEGE OF ARTS & SCIENCE

(Affiliated to Bharathiar University & Accredited by NAAC)
KGiSL Campus, Saravanampatti, Coimbatore- 641035.

B.Sc. Computer Science

CORE LAB-5- PROGRAMMING LAB – Linux and Shell Programming Lab

LIST OF PROGRAMS

1. Write a shell script to stimulate the file commands: rm, cp, cat, mv, cmp, wc, split, diff.
2. Write a shell script to show the following system configuration:
    a. currently logged user and his log name
    b. current shell , home directory , Operating System type , current Path setting , current working directory
    c. show currently logged number of users, show all available shells
    d. show CPU information like processor type , speed
    e. show memory information
3. Write a Shell Script to implement the following: pipes, Redirection and tee commands.
4. Write a shell script for displaying current date, user name, file listing and directories by getting user choice.
5. Write a shell script to implement the filter commands.
6. Write a shell script to remove the files which has file size as zero bytes.
7. Write a shell script to find the sum of the individual digits of a given number.
8. Write a shell script to find the greatest among the given set of numbers using command line arguments.
9. Write a shell script for palindrome checking.
10. Write a shell script to print the multiplication table of the given argument using for loop.

PREREQUISITES

Software and Pre require knowledge

Software Requirements:

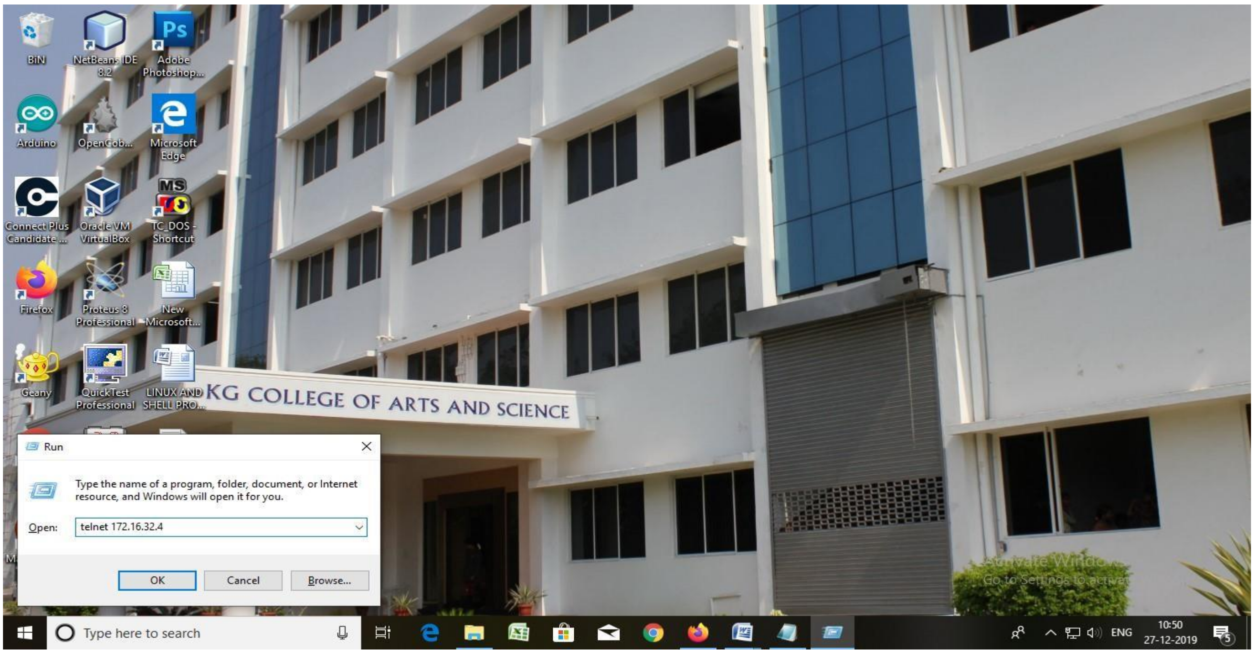
- Linux Operating System
- Telnet or Putty
- Browsers

Basic Skill Set Required:

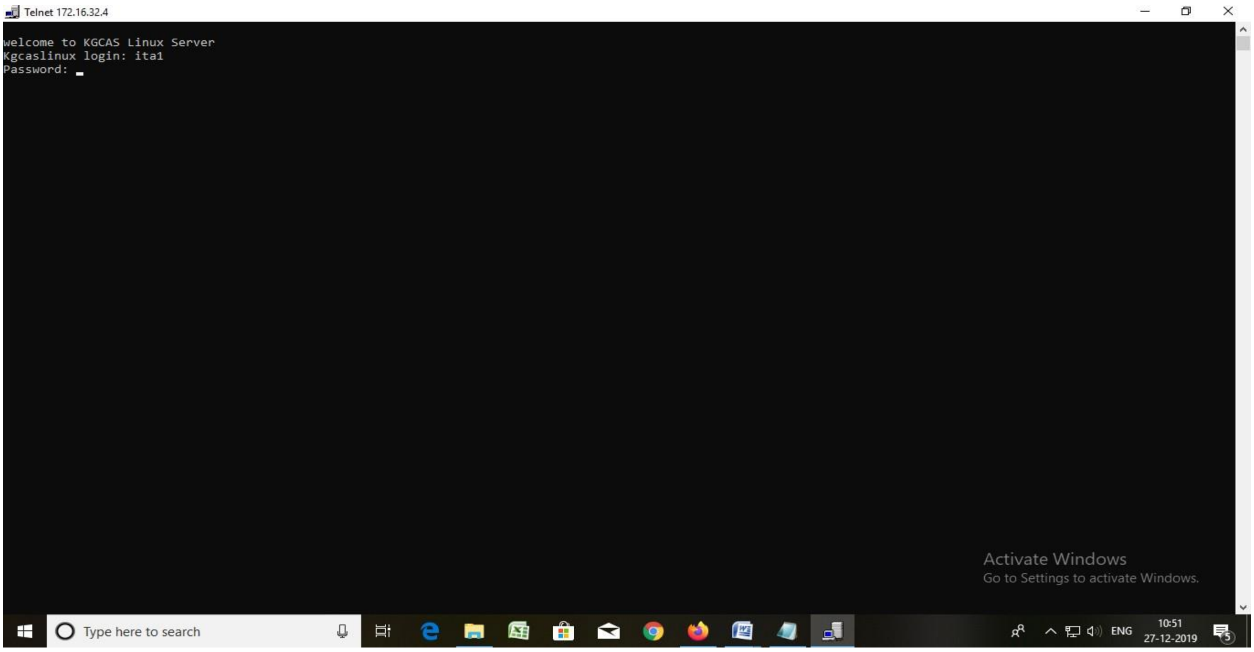
- To open a file and command prompt.
- Typing skills.
- Basic Knowledge about Command Line Interface (CLI).
- Fundamental concepts in C language.

GUIDE TO UTILIZE TOOL (MANUAL FOR TOOL)

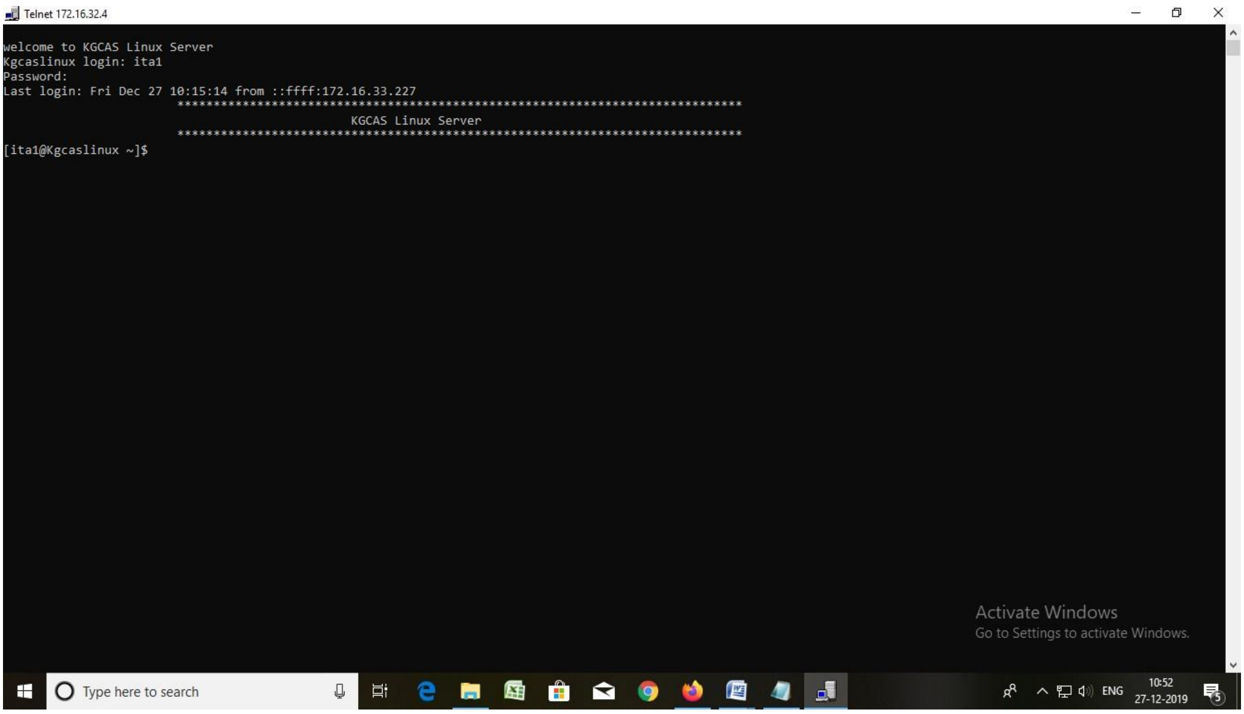
- **Steps for accessing the tool**
  1. Going to Run Command and access telnet by typing Telnet 172.16.32.4



  2. Type the Correct UserName and Password for logging to Linux



3. After Successful Logging it will connected Linux Server Machine.



ExNo: 01 **1**

FILE MANIPULATION COMMANDS

AIM:
To write a shell script to stimulate the basic Linux commands: rm, cmp, cat, cp, mv, wc, split, diff.

ALGORITHM:
Step1: Start the process.
Step2: In the shell script prompt window perform the required commands.
Step3: Use ls command to list the files and directories in the current directory.
Step4: Copy the contents of the file1 to file3, use cp command.
Step5: Use cat command to view the copied content in the file3.
Step6: Use mv command to move the contents in file2 to file4. And view the content in file4.
Step7 : Use wc command to display count of the lines, words, character in the file file4.txt
Step8: Use wc -l, wc -w, wc -c to display the count of lines, word, character respectively.
Step9: Use the split command to split the contents in the file4.
Step10: Use cmp command to compare the contents file1 and file4
Step11: Using diff command between file1.txt file4.txt
Step12: Stop the process.

SHELL SCRIPT:
```
echo “ ls command”
ls
echo "Copying content from file1.txt to file3.txt"
cp file1.txt file3.txt
echo "Displaying contents in file3.txt"
cat file3.txt
echo "Moving contents from file2.txt to file4.txt"
mv file2.txt file4.txt
echo "Displaying content in file4.txt"
cat file4.txt
echo "Display count of the lines, words, character in the file file4.txt"
wc file4.txt
echo "Display the number of lines in file4.txt"
wc -l file4.txt
echo "Display the number of words in file4.txt"
wc -w file4.txt
echo "Display the number of character in file4.txt"
wc -c file4.txt
echo "Splitting the file4.txt by 3 lines in each"
split -3 file4.txt
echo "Listing the files"
ls
echo "Comparing the files file1.txt file4.txt"
cmp file1.txt file4.txt
echo "Using diff command between file1.txt file4.txt"
diff file1.txt file4.txt
```

OUTPUT:

ls command
content.txt file3.txt new.txt program1.sh program2.sh program4.sh program5.sh progra
m7.sh xab
file1.txt file4.txt passwd program1.txt program3.sh program4.txt program6.sh xaa
xac
Copying content from file1.txt to file3.txt
Displaying contents in file3.txt
a
b
c
d
e
f
g
The appended content
Moving contents from file2.txt to file4.txt
Displaying content in file4.txt
sun
mon
tue
wed
thu
fri
sat
Display count of the lines, words, character in the file file4.txt
7 7 28 file4.txt
Display the number of lines in file4.txt
7 file4.txt
Display the number of words in file4.txt
7 file4.txt
Display the number of character in file4.txt
28 file4.txt
Splitting the file4.txt by 3 lines in each
Listing the files
content.txt file3.txt new.txt program1.sh program2.sh program4.sh program5.sh progra
m7.sh xab
file1.txt file4.txt passwd program1.txt program3.sh program4.txt program6.sh xaa
xac
Comparing the files file1.txt file4.txt
file1.txt file4.txt differ: byte 1, line 1
Using diff command between file1.txt file4.txt
1,8c1,7
< a
< b
< c
< d
< e
< f

```
< g
< The appended content
\ No newline at end of file
---
> sun
> mon
> tue
> wed
> thu
> fri
> sat
```

ExNo: 02 **4**

SYSTEM CONFIGURATION COMMANDS

AIM:
To write a shell script to implement the user and system information by commands.

ALORITHM:
Step1: Start the process.
Step2: In the shell script prompt window perform the following command: use LOGNAME to check login name of the user, SHELL to check the following shell information.
Step3: Type OSTYPE command to check the OS type of the Linux OS.
Step4: Type path to check the path for particular directories.
Step5: Type pwd command to view the present working directory.
Step6: lscpu command to check the CPU information.
Step7: Stop the process.

SHELL SCRIPT:
```
echo "User Name: " $USER
echo "Login Name: " $LOGNAME
echo "Current Shell: " $SHELL
echo "List of Shells: "
chsh -l
echo "Home Directory: " $HOME
echo "Our PC OS is: "$OSTYPE
echo "Current Path" $PATH
echo "Current Directory: "
pwd
echo "List of Logged Users"
who|wc -l
echo "System Configuration (or) PC Configuration: "
lscpu
echo "Free Memory Space: "
free –m
echo “Showing all Memory Information:”
cat/proc/meminfo
```

OUTPUT:
```
User Name: 1922kc38
Login Name: 1922kc38
Current Shell: /bin/bash
List of Shells:
/bin/sh
/bin/bash
/sbin/nologin
/usr/bin/sh
/usr/bin/bash
/usr/sbin/nologin
/bin/tcsh
/bin/csh
/bin/zsh
```

Home Directory: /home/1922kc38
Our PC OS is: linux-gnu
Current Path:
/usr/local/bin:/bin:/usr/bin:/usr/local/sbin:/usr/sbin:/home/1922kc38/.composer/vendor/bin:/usr/lib64/openm
pi/bin:/usr/bin:/home/1922kc38/.local/bin:/home/1922kc38/bin
Current Directory:
/home/1922kc38/linuxLab
List of Logged users:
43
System Configuration (or) PC Configuration:
Architecture: x86_64
CPU op-mode(s): 32-bit, 64-bit
Byte Order: Little Endian
CPU(s): 2
On-line CPU(s) list: 0,1
Thread(s) per core: 1
Core(s) per socket: 2
Socket(s): 1
Free memory space:

	total	used	free	shared	buff/cache	available
Mem:	3683	547	2634	33	501	2759
Swap:	3967	0	3967			

ExNo: 03 **6**

IMPLEMENTION OF PIPE, REDIRECTION AND TEE COMMANDS

AIM:

To write a shell script to implement the following pipe, redirection and tee commands.

ALGORITHM:

Step1: Start the process.
Step2: Use (|) pipe command to link one command with another (or) one operation.
Step3: Use (>>) command to transfer the file from one part into another.
Step4: Use more command to check the information on the screen.
Step5: Display the result on the screen.
Step6: Save the process.
Step7: End the process.

SHELL SCRIPT:

```
echo "Exploring Pipes and Redirection Commands"
echo "Using pipe command:"
ls –l | wc
echo "Using tee command:"
ls –l | wc | tee new.txt
echo "Content in new.txt"
cat new.txt
echo "Using redirection command:"
echo "Content in file1.txt before redirecting:"
cat file1.txt
echo "Redirecting commands is to be executed… "
cat >> file1.txt
echo "The content in file1.txt after appending:"
cat file1.txt
```

OUTPUT:

```
Exploring pipes and Redirection Commands
Using pipe command:
19     164    1107
Using tee command:
19     164    1107
Content in new.txt
19     164    1107
Using redirection command:
Content in file1.txt before redirecting:
a
b
c
d
e
f
g
The appended content
Redirecting commands is to be executed....
```

this is the appended text
The content in file1.txt after appending:
a
b
c
d
e
f
g
The appended content
this is the appended text

ExNo: 04 **8**

SHELL SCRIPT FOR DISPLAYING DATE, USERNAME AND LISTING THE FILES AND DIRECTORIES

AIM:
To write a shell script to display the current date, username and list of files and directories by getting users choice.

ALGORITHM:
Step 1: Start the process.
Step 2: Use a case statement for performing a different action into a single prompt.
Step 3: Declare case variable and case command for the program.
Step 4: If the choice is 1 then the current date will be displayed on the screen.
Step 5: If the choice is 2 than username will be shown if the choice is 3 then file can be listed along with the directories.
Step 6: If name of the choice is met finally default case get executed on the screen.
Step 7: Stop the process

SHELL SCRIPT:
```
echo "1.Current date:"
echo "2.Your user name:"
echo "3.List files and directories"
read option
case ${option} in
1)
echo "Current date is :" $(date);;
2)
echo "Your user name is: "$(whoami);;
3)
echo "To list out all files and directories:"$(ls);;
*)
echo "Invalid Option"
esac
```

OUTPUT:
```
1. Current Date:
2. Your user name:
3. List files and directories
1
Current date is: Fri Feb 26 15:58:15 IST 2021
```

ExNo: 5

IMPLEMENTATION OF FILTER COMMANDS

AIM:
To write a shell script to implement filter commands.

ALGORITHM:
Step 1: Start the process.
Step 2: Create a file using vi editor.
Step 3: Copy the file /etc/passwd to passwd file.
Step 4: To display the lines containing the word root use grep -n "root" passwd.
Step 5: To display the no.of lines containing the word root use grep -c "root"passwd.
Step 6: To display all lines, words, characters in passwd file use wc passwd.
Step 7: To display all the lines that do not match with the line root use grep -v"root "passwd.
Step 8: To replace ":"with "*"in the file passwd use tr":" "*"<passwd.
Step 9: To display the first column of the file passwd use cut -d ':' -f1 passwd.
Step 10: The output will be displayed on the screen.
Step 11: Stop the process.

SHELL SCRIPT:
```
echo "Filter commands"
cp /etc/passwd passwd
echo "Display the lines containing the word root"
grep -n "root" passwd | more
echo "Display the count of lines that is containing the word root"
grep -c "root" passwd
echo "Display the count of lines that dont match with the line root"
grep -v "root" passwd | more
echo "Display the no. of lines, character and words in passwd file"
wc passwd
echo "Replace ":" with "*" in the passwd file"
tr ':' '*' < passwd | more
echo "Display first column of the passwd file"
cut -d ':' -f1 passwd
```

OUTPUT:
```
[1922kc62@Kgcaslinux lab]$ sh program5.sh
Filter commands
Display the lines containing the word root
1:root:x:0:0:root:/root:/bin/bash
10:operator:x:11:0:operator:/root:/sbin/nologin
Display the count of lines that is containing the word root
2
Display the count of lines that dont match with the line root
bin:x:1:1:bin:/bin:/sbin/nologin
daemon:x:2:2:daemon:/sbin:/sbin/nologin
adm:x:3:4:adm:/var/adm:/sbin/nologin
lp:x:4:7:lp:/var/spool/lpd:/sbin/nologin
sync:x:5:0:sync:/sbin:/bin/sync
Display the no. of lines, character and words in passwd file
```

1626 1670 76018 passwd
Replace : with * in the passwd file
root*x*0*0*root*/root*/bin/bash
bin*x*1*1*bin*/bin*/sbin/nologin
daemon*x*2*2*daemon*/sbin*/sbin/nologin
adm*x*3*4*adm*/var/adm*/sbin/nologin
lp*x*4*7*lp*/var/spool/lpd*/sbin/nologin
sync*x*5*0*sync*/sbin*/bin/sync
shutdown*x*6*0*shutdown*/sbin*/sbin/shutdown

ExNo: 06

DELETE THE FILE WHICH HAS SIZE AS ZERO FILE SIZE

AIM:
To write a shell script to remove the file which has size as zero bytes.

ALGORITHM:
Step 1: Start the process.
Step 2: create a file using vi editor.
Step 3: Get a file name as input from the user.
Step 4: Use the if ..... else statement check the condition.
Step 5: If the file exists then check the size of the file.
Step 6: If the size of file is zero then remove the file.
Step 7: Remove the file using rm command .if not leave it as it is.
Step 8: Stop the process.

SHELL SCRIPT:
```
echo "Enter the filename:"
read fnm
if [ -e $fnm ]
then
echo $fnm" file exist"
if [ -s $fnm ]
then
echo $fnm" file has size > 0"
else
rm $fnm
echo $fnm" file is deleted which has size = 0"
fi
else
echo "File does not exist"
fi
```

OUTPUT:
```
[1922kc62@Kgcaslinux lab]$ sh program6.sh
Enter the filename:
content.txt
content.txt file exist
content.txt file has size > 0
```

ExNo: 07 **12**

FINDING THE GREATEST AMONG THE GIVEN NUMBERS USING COMMAND LINE ARGUMENTS

AIM:
To write a shell script to find the greatest among the given number using command line argument.

ALGORITHM:
Step1: Start the process.
Step2: Create a file using vi editor.
Step3: Using echo print the statement.
Step4: Get the file name as input from the user.
Step5: Using the for loop check the condition and print the numbers stored in the array.
Step6: Using if print the statement as greater and smaller numbers.
Step7: Print the smallest numbers and largest numbers.
Step8: The output will be displayed on the screen.
Step9: Read the total number count for an array can be get from user using command line arguments.
Step10: Stop the process.

SHELL SCRIPT:
```
for((i=0;i<$1;i++))
do
echo "Enter $((i+1)) number:"
read nos[$i]
done
echo "Number entered are:"
for((i=0;i<$1;i++))
do
echo ${nos[$i]}
done
small=${nos[0]}
greater=${nos[0]}
for((i=0;i<$1;i++))
do
if [ ${nos[$i]} -lt $small ];
then
small=${nos[$i]}
elif [ ${nos[$i]} -gt $greater ];
then
greater=${nos[$i]}
fi
done
echo "Smallest number in an array is $small"
echo "Greatest number in an array is $greater"
```

OUTPUT:

```
[1922kc62@Kgcaslinux lab]$ sh program8.sh 3
Enter 1 number
6
Enter  2 number
3
Enter  3 number
9
Numbers entered are:
6
3
9
Smallest number in an array is 3
Greatest number in an array is 9
```

ExNo: 08

FINDING THE SUM OF INDIVIDUAL DIGITS OF GIVEN NUMBER

AIM:
To write a shell script to find the sum of individual digits.

ALGORITHM:
Step 1: Start the process**.**
Step 2: Create a file using vi editor.
Step 3: Using echo print the statement “Enter the number”.
Step 4: Read the value from user whose summation of individual digits can be found.
Step 5: Declare & initialize the variables sd=0 and sum=0.
Step 6: Find n%10,n/10,sum=sum + sd and store it in sd, n, sum respectively.
Step 5: Repeat the step 6 until n greater than 0 using while loop.
Step 6: Display the output.
Step 7: Stop the process

SHELL SCRIPTS:
```
echo -n "Enter a number:"
read n
sd=0
sum=0
while [ $n -gt 0 ]
do
sd=$(( $n % 10 ))
n=$(( $n / 10 ))
sum=$(( $sum + $sd ))
done
echo "Sum of all digits is "$sum
```

OUTPUT:
```
[1922kc62@Kgcaslinux lab]$ sh program7.sh
Enter a number: 1234
Sum of all digits is 10
```

ExNo: 09

SHELL SCRIPT FOR CECKING THE GIVEN STRING OR NUMBER IS PALINDROME OR NOT

AIM:
To write a shell script for palindrome checking.

ALGORITHM:
Step 1: Start the process.
Step 2: Get the string from the user.
Step 3: Reverse the string using rev<<<string
Step 4: If the reversed string and the given string is same display it as palindrome
Step 5: Else display not a palindrome for this use if else.
Step 6: Stop the process

SHELL SCRIPT:
```
read -p "Enter a string:" string
if [[ $(rev <<< "$string") == "$string" ]];
then
echo "palindrome"
else
echo "not a palindrome"
fi
```

OUTPUT:
Enter a string: 1221
Number is palindrome

ExNo: 10

PRINTING THE MULTIPLICATION TABLE USING FOR LOOP

AIM:
To write a shell script to print multiplication tables.

ALGORITHM:
Step 1: Start the process.
Step 2: Get the table number and range from the user.
Step 3: Use for loop to find the table of the given number and range.
Step 4: Display the calculated table in the standard table format.
Step 5: Stop the process

SHELL SCRPIT:
```
echo "Enter the table number"
read n
echo "Enter the range"
read range
echo "Multiplication table for $n upto the range $range"
for((i=1;i<=range;i++))
{
echo " $i X $n = `expr $n \* $i`"
}
```

OUTPUT:
```
Enter the table number
1
Enter the range
4
Multiplication table for 1 upto the range 4
1 X 1 = 1
2 X 1 = 2
3 X 1 = 3
4 X 1 = 4
```